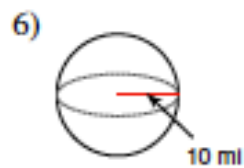
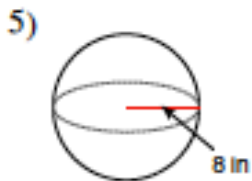
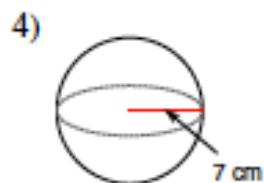
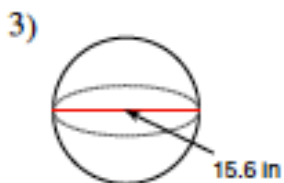
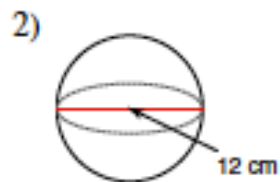
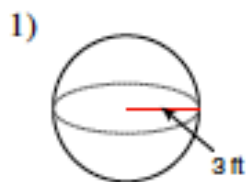


8.6/8.7 Volume and Surface Area of Spheres Worksheet

MPM1D

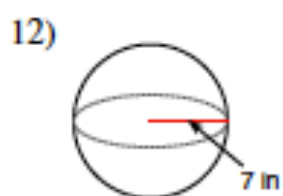
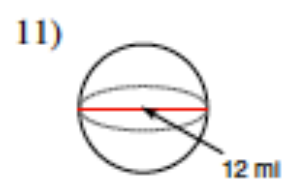
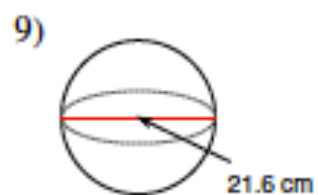
Part A: Find the Surface Area of Each Sphere



7) A sphere with a diameter of 6.2 in.

8) A sphere with a radius of 10 mi.

Part B: Find the Volume of Each Sphere



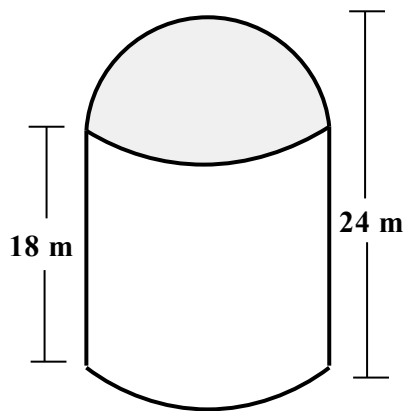
15) A sphere with a diameter of 2 m.

16) A sphere with a diameter of 10 ft.

Part C: Applications

17) Farmers store wheat in silos to protect the harvest from the weather.

- Determine how much paint is required to paint the surfaces of the silo that are exposed to the weather.
- Determine how much wheat the silo can hold?



1L covers 13m^2

18) A basketball has a diameter of 24.8 cm.

- How much leather is required to cover this ball, to the nearest tenth of a square centimeter?



- If the leather costs $\$28/\text{m}^2$, what does it cost to cover the basketball?

19) Emily is placing a gazing ball in one of her customer's gardens. The ball has a diameter of 60 cm and will be covered with reflective crystals. One jar of these crystals covers 1 m^2 .

a) Estimate the surface area to decide whether one jar of the crystals will cover the ball.

b) Calculate the surface area, to the nearest square centimeter.



20) Golf balls are stacked three high in a rectangular prism package. The diameter of one ball is 4.3 cm. What is the minimum amount of material needed to make the box?



Answers:

1) 113.1 ft^2 2) 452.4 cm^2 3) 764.5 in^2 4) 615.8 cm^2 5) 804.2 in^2 6) 1256.6 mi^2
 7) 120.8 in^2 8) 1256.6 mi^2 9) 5276.7 cm^3 10) 2144.7 in^3 11) 904.8 mi^3 12) 1436.8 in^3
 13) 7238.2 in^3 14) 113.1 mi^3 15) 4.2 m^3 16) 523.6 ft^3 17) a) 69.6L b) 2488.2 m^3
 18) a) 0.19 m^2 b) \$5.32 19) a) answers may vary b) $\text{SA} = 1.13 \text{ m}^2$ 20) 258.86 cm^2